

PROJECT CHARTER

[Project Name — Shishu-Sneh]

MindMatrix Industry Readiness Programme · Android Internship

Document Control

Document Title	Shishu-Sneh (Healthcare)
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Prepared By	[mohammed jakeer f maniyar] · Project Manager
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Last Revised	[27 April 2026] — [Brief change description]
PRD Reference	[Shishu-Sneh PRD v1.0] — attached as Annex A

1. Project Overview

1.1

1.2 Project Identity

Project Name	[Android App Development using GenAI- Shishu-Sneh (Healthcare)]
Project Code	Shishu-Sneh PRD v1.0
Project Type	Android Application Development · 6-Week Internship Deliverable
Tech Stack	Kotlin · Jetpack Compose · Room Database · Firebase · MVVM Architecture
PRD Section Reference	All section references in this Charter refer to the attached PRD (Annex A)

1.3 Project Objective

To develop a mobile application that assists new mothers in tracking baby growth, vaccination schedules, and developmental milestones during the first year.

1.4 Problem Statement

New mothers, especially in rural areas, often discontinue exclusive breastfeeding early due to lack of awareness, social myths, or work-related pressures. Additionally, after leaving the hospital, they lack proper tools to track their baby's weight, monitor growth milestones, and manage vaccination schedules. This results in missed immunizations, improper nutrition, and increased health risks for infants.

1.5 Success Vision

Shishu-Sneh acts as a “Baby’s First Year Guide,” functioning like a digital elder for new mothers. The application provides:

- Weekly developmental milestone tracking (e.g., baby smiling, head control)*
- A simple growth chart to monitor weight and height trends*
- A vaccination calendar with alerts and disease awareness*
- Daily feeding and nutrition guidance for mother and child*

The goal is to provide accessible, reliable, and easy-to-understand healthcare support in a digital format.

2. Stakeholders

2.1 Project Sponsor

Name	<i>Mr. Tirumal Mutalikdesai</i>
Role	Project coordinator— approves scope, accepts deliverables, scores evaluation
Organisation	MindMatrix
Contact	<i>[program_coordinator@mindmatrix.io · 9611546444]</i>
Availability	<i>[Available Mondays and Thursdays 10am–12pm for queries]</i>
Decision Authority	Approves Project Charter, accepts Change Requests, signs off final deliverables

2.2 Project Team — Roles (From PRD Section 17)

Role (PRD §17)	Team Member Name	Primary Responsibility — What They Own	Deliverable Ownership
Project Manager / Team Lead	[MohammedJ akeer F Maniyar]	Project planning, documentation, milestone tracking, risk management, and coordination of all modules	Project Report (§14.5), Lessons Learned
UI Developer	[MohammedJ akeer F Maniyar]	Design and development of all Jetpack Compose screens including Growth Chart, Vaccination Alerts, Feeding Guide, and Milestone Checklist. Handles navigation flow and UI consistency	Wireframe (§14.1), Prototype (§14.3), Video Demo (§14.6)
Database Handler	[MohammedJ akeer F Maniyar]	Design and implementation of Room Database including entities (Baby, Vaccination, Growth), DAOs, and repository pattern. Ensures data consistency and privacy	DB Schema Diagram (EPT §14.2), Room Database Integration
Feature Integrator	[MohammedJ akeer F Maniyar]	Integration of growth charts (MPAndroidChart), vaccination reminders using WorkManager, and core feature logic implementation	Final APK (§14.4), Feature Integration
Analytics Developer (EPT)	[MohammedJ akeer F Maniyar]	Implementation of growth trend analysis using charts and data visualization for baby health tracking	Analytics Dashboard Screenshots (§14.4)

2.3 External Stakeholders

Stakeholder	Interest / Power	Communication Frequency	Notes
MindMatrix Programme Evaluator	HIGH POWER: Evaluates project based on functionality, UI/UX, and documentation. Final authority for grading	Week 6: Submission of all deliverables and final evaluation	Ensure all required documents (SRD, Charter, APK, Report, Demo Video) are complete and aligned with evaluation criteria
End Users — Shishu-Sneh (Healthcare) New Mothers (Primary Users)	LOW POWER, HIGH INTEREST: They are the main users of the app and validate usability and usefulness	Week 5: User testing (UAT) or demo-based validation	Ensure the app is simple, easy to use, and understandable for non-technical users, especially rural mothers

End Users — Healthcare Workers / ASHA Workers	MEDIUM POWER, HIGH INTEREST: Can recommend or guide mothers in using the app	Week 5: Feedback and usability validation	Ensure vaccination reminders and health data are accurate and helpful
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3. Project Scope

3.1 In-Scope Features (From PRD Section 3)

Feature ID	Feature Name (exact from PRD §3)	PRD §5 Section	Priority
F-01	Growth Chart	§3 App Usage	Must Have
F-02	Vaccination Alert	§3 App Usage	Must Have
F-03	Feeding Guide	§3 App Usage	Must Have
F-04	Milestone Checklist	§3 App Usage	Must Have
F-05	Weight Tracker	§2 Vision	Must Have
F-06	Immunization Calendar	§2 Vision	Must Have
F-07	Growth Trend Line Visualization	§4 Technical Implementation	Should Have
F-08	Weekly Developmental Milestones	§2 Vision	Could Have

3.2 Out-of-Scope Exclusions (From PRD Section 3)

Exclusion (from PRD §3)	Rationale	Future Scope Recommendation (§15)
Real-time doctor consultation / Telemedicine	Requires backend infrastructure, medical compliance, and certified professionals	Add telemedicine feature in Phase 2 with verified doctors and secure video consultation
AI-based disease prediction for babies	Requires machine learning models, datasets, and medical validation	Integrate Gemini AI or healthcare ML models in future version
Cloud data sync across multiple devices	Requires backend server and internet dependency	Implement Firebase Cloud Sync in Phase 2

Multi-user login (family sharing)	Adds complexity in authentication and data privacy handling	Add multi-user support with Firebase Authentication later
Advanced analytics & medical reports	Requires complex data processing and medical interpretation	Enhance analytics dashboard with detailed insights in Phase 2
integration with government health systems	Requires API access and regulatory permissions	Integrate with health APIs (like Ayushman Bharat) in future
voice assistant / speech-based guidance	Requires NLP and speech processing integration	Add voice assistant using AI APIs in Phase 2

3.3 Assumptions

Technology availability: Android Studio, Kotlin, Jetpack Compose, Room DB, WorkManager, and MPAndroidChart libraries are available and properly configured from Week 1.

PRD stability: The project requirements defined in the Shishu-Sneh PRD are final. No major feature additions or changes will be made without a formal Change Request.

Mentor availability: The mentor will be available at least once a week for guidance, reviews, and feedback on project progress.

Library compatibility: MPAndroidChart is compatible with Jetpack Compose and works correctly for displaying baby growth trends. (To be tested in Week 1)

Offline-first functionality: The application will primarily work offline using Room Database. Internet is not required for core features like growth tracking and milestone checking.

User data accuracy: Users (mothers) will enter correct baby details such as weight, height, and date of birth for accurate tracking and reminders.

Device compatibility: The app will run smoothly on Android devices with minimum API Level 28 (Android 9.0) and above.

Notification reliability: WorkManager will successfully trigger vaccination reminders based on the baby's date of birth.

Single-user usage: The application is designed for one user (mother) per device; multi-user functionality is not considered in this version.

3.4 Constraints

Timeline: The project must be completed within 6 weeks as per PRD §12. Final submission will be in Week 6 with no extensions allowed.

Technology: The application will be developed only for Android using Kotlin, Jetpack Compose, Room Database, and WorkManager as defined in PRD §7.

Compatibility: The application must support Android 9.0 (API Level 28) and above to ensure wider device usability.

Team Size: The project will be developed by **1 member (individual project)**. No additional team members will be added.

Budget: The project must be completed with zero external cost. Only free tools, open-source libraries, and student resources will be used.

Offline Functionality Constraint: The app must work primarily offline using Room Database. Internet-based features will be minimal.

Device Limitation: Testing will be performed using emulator and limited real devices, which may restrict performance testing across all device types.

4. Deliverables

4.1 Project Deliverables (From PRD Section 14)

ID	Deliverable (PRD §14)	Owner	Due	Acceptance Condition
D- 01	Wireframe (Figma/Canva) — all §10 screens with colour palette	UI Developer	End Week 1	All key screens (Growth Chart, Vaccination Alert, Feeding Guide, Milestone Checklist) are designed. Mentor approval received. Soft and mother-friendly UI maintained.
D- 02	User Flow Diagram / DB Schema ER Diagram	UI Dev / DB Handler	End Week 2	User flow clearly shows app navigation (Home → Features → Data Entry → Results). Database schema includes Baby Profile, Growth Records, Vaccination Records with proper relationships. Mentor reviewed.
D- 03	Functional UI Prototype — all screens navigable, mock data	UI Developer	End Week 2	All screens open properly with navigation working. Mock data is displayed. No crashes during demo.
D-04	Final APK (Working App)	Feature Integrator	End Week 5	App installs successfully. Growth chart, vaccination reminders, and milestone tracking are functional. No major bugs.
D-05	Database Implementation (Room DB)	Database Handler	End Week 4	Data is stored and retrieved correctly. Offline functionality works. No data loss after app restart.
D-06	Testing Report (Bug List + Fixes)	Project Manager	End Week 5	At least 5 bugs identified and resolved. App tested for edge cases (empty input, invalid data, restart).

Programme D-07	Final Documentation (README + Report)	Project Manager	Week 6	Complete documentation including problem statement, features, tech stack, and screenshots submitted.
D-08	Demo Video	UI Developer	Week 6	Video shows full app flow (start → features → output). Clear explanation of project.

4.2 PM Artefacts (Not in PRD §14 — Required for Professional Delivery)

Artefact	Owner	Due	Purpose
This Project Charter	Team Lead (Mohammed Jakeer F Maniyar)	Week 1	Defines project scope, objectives, constraints, and assumptions. Acts as a baseline document for the entire project.
Work Breakdown Structure (WBS)	Team Lead (Mohammed Jakeer F Maniyar)	Week 1	Breaks the project into smaller tasks (UI, Database, Features) for proper planning and task assignment.
6-Week Gantt Chart	Team Lead (Mohammed Jakeer F Maniyar)	Week 1	Provides timeline of project phases (Design, Development, Testing, Deployment) with milestones tracking.
Risk Register (min. 10 risks)	Team Lead (Mohammed Jakeer F Maniyar)	Week 1	Identifies potential risks like app crashes, data loss, or delays and plans mitigation strategies.
Requirements Traceability Matrix	All	Week 2	Maps features (Growth Chart, Vaccination Alert, etc.) to design, implementation, and testing for full traceability.
Weekly Status Updates ×6	Team Lead	Each Friday	Tracks weekly progress (Completed, Planned, Issues) and ensures project stays on schedule.
Lessons Learned Report	Team Lead	Week 6	Documents challenges faced, solutions applied, and key learnings for future improvement.

5. Milestone Schedule (From PRD Section 12)

Milestone	Section 12 Description	Target Date	Demo / Evidence	Status
◆ M1 — Week 1	Setup & Wireframes — skeleton, folder structure, wireframes approved	22 April 2026	Mentor written approval on wireframes	Not Started
M2 — Week 2	UI Development — all Compose screens navigable with mock data	29 April 2026	Live screen demo: all screens reachable, no crashes	Not Started
M3 — Week 3	Database Integration — Room CRUD all entities tested	06 May 2026	CRUD operations shown in Logcat to mentor	Not Started
M4 — Week 4	Feature Integration — growth chart + vaccination alerts working	13 May 2026	Live demo: data entry → chart update → alert trigger	Not Started
M5 — Week 5	Testing & Refinement — UI polished, app stable	20 May 2026	Performance check + smooth UI on emulator	Not Started
M6 — Week 6	Demo & Report — final submission of all deliverables	27 May 2026	APK + Report + Video Demo + Final Acceptance	Not Started

6. Risk Register — Executive Summary

ID	Risk Description	P	I	Score	Zone	Response	Owner
R-01	MPAndroidChart not compatible with Jetpack Compose	4	5	20	 RED	Mitigate: Test chart integration in Week 1	Feature Integrator
R-02	Room Database schema changes after Week 2 freeze	4	4	16	 RED	Mitigate: Finalize DB schema early and review	DB Handler
R-03	WorkManager notifications not triggering properly	3	4	12	 AMBER	Mitigate: Test reminders with different dates in Week 2	Feature Int.
R-04	Incorrect user input (wrong baby data) affects results	3	3	9	 AMBER	Mitigate: Add input validation and error messages	UI Developer
R-05	App crashes due to null values (NPE errors)	4	4	16	 RED	Mitigate: Use safe calls and proper null handling	Developer
R-06	Scoop creep (adding extra features like AI,login)	4	4	16	 RED	Avoid:Stick strictly to defined features in PRD	Team Lead
R-07	UI not user-friendly for rural mothers	3	4	12	 AMBER	Mitigate: Keep UI simple, test with basic users	UI Developer
R-08	Data loss due to improper Room DB handling	3	5	15	 RED	Mitigate: Test data persistence and backup logic	Database Handler
R-09	Performance issues on low-end devices	3	3	9	 AMBER	Mitigate: Optimize UI and avoid heavy operations	Developer
R-10	Delay in documentation and final submission	5	4	20	 RED	Mitigate:Prepare documentation parallely	Team Lead

7. Resource Budget — 180 Productive Hours

Component	PRD Reference	Estimated Hours	Owner Role	Notes
Jetpack Compose UI — all screens	§10 Key Screens	~35 hrs	UI Developer	4 main screens (Growth, Vaccine, Feeding, Milestone)
Room DB — entities, DAOs, Repository	§7 · §8 Schema	~30 hrs	DB Handler	Baby, Growth, Vaccination entities + CRUD
MVVM ViewModels + StateFlow	§7 Architecture	~15 hrs	Feature Integrator	Basic state handling for all features
WorkManager (Notifications)	§3 In-Scope · §7	~15 hrs	Feature Integrator	Vaccination reminders setup
Charts (MPAndroidChart)	§11 Analytics	~15 hrs	Feature Integrator	Only growth line chart (keep simple)
Testing (Functional + NFR)	§6 · §16 (30%)	~20 hrs	All	Emulator + edge case testing
Documentation (Report + Video)	§14 Deliverables	~25 hrs	Team Lead	Report + demo video + screenshots
WBS / Gantt / Charter	PM	~10 hrs	Team Lead	Planning documents
Project Management overhead	PM updates	~10 hrs	Team Lead	Weekly updates
Buffer (unexpected issues)	Risk reserve	~5 hrs	All	Use only if needed

8. Acceptance Criteria

8.1 Evaluation Criteria (From PRD Section 16)

Criterion	Weight	Acceptance Standard	How to Test
Functionality	30%	All §3 Must Have features (Growth Tracker, Vaccination Alerts, Feeding Guide, Milestone Checklist) work end-to-end using Room DB. No crashes in main flows.	Test each feature: Add baby weight → verify chart updates. Set vaccination → verify reminder triggers. Mark milestone → verify saved state. Check app restart persistence.
UI / UX Design	25%	All screens follow soft, mother-friendly UI. Navigation matches defined user flow. Easy to use without guidance.	Compare screens with wireframe. Perform usability test: Can a user navigate Growth, Vaccine, Feeding without help? Check font size, colors, spacing consistency.
Code Quality	20%	MVVM architecture followed. No direct Room DB calls in UI. Repository pattern used.	Review code: Ensure Composables don't access DAO directly. Check ViewModel uses StateFlow. Verify clean separation (UI → ViewModel → Repository → DB).
Innovation	15%	At least one extra feature implemented (e.g., Local language support / smart tips).	Demonstrate feature: Show feeding tips or multilingual support working in app. Mention clearly in report under Innovation section.
Presentation	10%	Report follows PRD structure. Video demo explains features clearly within 2–3 minutes.	Check report includes PRD references (§3, §5, §10). Record demo: Show Growth, Vaccination, Milestones flow clearly within time limit.

8.2 Non-Functional Acceptance Tests (From PRD Section 6)

NFR ID	PRD	Criterion	Test Procedure	Pass Threshold
NFR-PER-F-01	Performance	Growth chart & screens load within 2 seconds	Open Growth Chart with 20+ records. Measure using Android Profiler (CPU time). Run 3 times and take average.	< 2000 ms average
NFR-POR-T-01	Portability	App compatible with Android 9.0+	Run full app on API 28 emulator. Test all features (Growth, Vaccination, Milestones).	Zero crashes, all features working
NFR-REL-Y-01	Reliability	No data loss after app restart	Add baby data (weight, milestones), force close app, reopen app and verify data	100% data retained
NFR-SEC-01	Security	Baby health data stored securely	Inspect Room DB file in emulator (Device Explorer). Check if data is exposed directly	Data not easily readable / protected
NFR-MAINT-01	Maintainability	Proper MVVM architecture followed	Code review: Check no DAO usage in UI layer, ViewModel uses StateFlow	0 DAO calls in UI layer

9. Communication Plan

Communication	Audience	Frequency	Channel	Content	Owner
Weekly Status Update	Mentor	Every Friday by 5pm	9611546444/program_coordinator@mindmatrix.io	Completed work, next plan, blockers, decisions needed (max 200 words)	Team Lead (Mohammed Jakeer)
Milestone Demo — Week 2	Mentor	End of Week 2	Google Meet (Video Call)	20-min live demo of all Compose screens with navigation. Compare with wireframes and take feedback	UI Developer
Milestone Demo — Week 4 (Critical)	Mentor + Evaluator	End of Week 4	Google Meet / In-person	Live demo with Room DB data (Growth chart, Vaccination alerts). Show Risk Register updates. No mock data	Full Team
Daily Team Standup	All Team Members	Every working day	WhatsApp Group	What done, what doing, blockers (max 5 minutes discussion)	All Members (rotating)

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Risk Alert	Mentor	Within 24 hrs of RED risk	<i>WhatsApp (Direct Message)</i>	Problem situation, impact, what tried, what help needed	Risk Owner
Formal Closeout — Week 6	Mentor + Evaluator	Week 6 (scheduled in Week 5)	<i>Google Meet / In-person</i>	Final APK demo, all deliverables submission, lessons learned, final approval	Final APK demo, all deliverables submission, lessons learned, final approval

10. Change Control Process

10.1 Rule

No changes will be made to the project scope without a documented and mentor-approved Change Request (CR).

10.2 Process — 5 Steps

1. IDENTIFY

Any new feature request or modification (e.g., adding AI-based baby health suggestions or multilingual support) must be documented in a Change Request (CR) form.
Assign a unique CR ID and record the date of request.

2. ASSESS

Evaluate the impact of the requested change:

- Affected milestone (PRD §12 timeline)
- Changes in existing features (PRD §3)
- Estimated additional development hours
- New risks introduced (e.g., API complexity, UI changes)

3. . DECIDE

Present the CR to the Project Mentor (Sponsor).

Decision will be one of the following:

- Approve → Implement in current phase
- Reject → Not feasible
- Defer → Move to Future Scope (§15)

Written approval (WhatsApp/email confirmation) is mandatory.

4. IMPLEMENT

If approved:

- Update Project Charter
- Update WBS & Gantt Chart
- Update Risk Register

Only after documentation update, development will begin.

10.3 Change Request Log

CR ID	Date	Description	Requested By	Impact (hrs + milestone)	Decision	Date Decided
CR - 001	[20-04-2026]	Add push notification reminders for vaccination instead of only in-app alerts	Mohammed Jakeer	+8 hrs. Affects Week 5 testing and notification setup	☐ Approved ☐ Rejected ☐ Deferred \$15	[25 april 2026]
CR - 002	[Date]	[Describe change request]	Team Member	+10 hrs. Affects Week 4 UI and Week 5 testing	☐ Approved ☐ Rejected ☐ Deferred \$15	[25 april 2026]
CR - 003	[Date]	Add baby growth comparison with standard WHO growth chart	DB Handler	+12 hrs. Affects Week 4 analytics and chart integration	☐ Approved ☐ Rejected ☐ Deferred \$15	[25 april 2026]

11. Charter Sign-Off

BY SIGNING BELOW, ALL PARTIES CONFIRM UNDERSTANDING AND ACCEPTANCE OF THIS PROJECT CHARTER AS THE FORMAL SCOPE BASELINE. ANY SUBSEQUENT CHANGE REQUIRES A FORMAL CHANGE REQUEST PER SECTION 10.

Role	Name	Signature	Date Signed
Project Manager / Team Lead	Mohammed Jakeer F Maniyar		[DD/MM/YYYY]
Product Owner / Mentor	[Tirumal mutalikdesai]		[DD/MM/YYYY]

Programme

UI Developer	Mohammed Jakeer F Maniyar		[DD/MM/YYYY]
Database Handler	Mohammed Jakeer F Maniyar		[DD/MM/YYYY]
Feature Integrator / Analytics	Mohammed Jakeer F Maniyar		[DD/MM/YYYY]
MindMatrix Programme Coordinator	Tirumal mutalik		[DD/MM/YYYY]

Sign-off Meeting Notes

The project "Shishu-Sneh – Baby's First Year Guide" is approved with the following conditions:

- The Minimum Viable Product (MVP) will include Growth Tracker, Vaccination Alerts, Feeding Guide, and Milestone Checklist.
- The application will follow an offline-first approach using Room Database. Firebase will be used only for authentication if time permits.
- Advanced features such as multilingual support and AI-based recommendations are considered future scope (§15).
- UI must be simple, clean, and mother-friendly as per usability goals.
- All features must be completed within the 6-week timeline without scope expansion.

Any additional feature beyond the defined scope must follow the Change Request process.

12. Glossary

Term	Definition
Project Charter	Formal document authorising the project. Defines scope, stakeholders, constraints, and acceptance criteria. Becomes the scope baseline once signed
PRD	Product Requirements Document — the source document (Shishu-Sneh PRD) provided by MindMatrix.
FR (Functional Requirement)	A specific behaviour the system must exhibit. Format: “FR-XX: The system shall...” Every FR is testable.
NFR (Non-Functional Requirement)	A quality constraint on how the system performs (performance, security, compatibility, etc.).
Scope Baseline	The signed and approved scope of the project. Any change requires a Change Request.
MVP (Minimum Viable Product)	The minimum features required (Growth, Vaccination, Feeding, Milestones) to make the app usable.
MVVM	Model-View-ViewModel — architecture used to separate UI, logic, and data.
Change Request (CR)	A formal request to change scope, timeline, or features. Requires mentor approval.
RTM	Requirements Traceability Matrix — mapping between requirements, design, and test cases.
WBS	Work Breakdown Structure — breakdown of tasks for project execution.